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DEPARTMENT OF DEFENSE HANDBOOK

ELECTROMAGNETIC ENVIRONMENTAL EFFECTS TO ORDNANCE GUIDE

PART 1

GENERAL GUIDANCE

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FOREWORD

1. This handbook is approved for use by all Departments or Agencies of the Department of Defense (DoD).

2. This handbook was prepared for all contractors, DoD test agencies, activities, and test planners that conduct Electromagnetic Environmental Effects (E3) Ordnance verification testing for airborne, sea, space, and ground systems and the associated ordnance. This handbook is for guidance only and cannot be cited as a requirement.

3. This handbook provides recommended practices for conducting E3 to Ordnance evaluations for compliance to military standards and Services' policy documents. Developed over many years by DoD test activities, the methods and procedures described herein are the product of the collective DoD ordnance test experience. This handbook consolidates these practices into a single document vice numerous individual Service standards, instructions, directives, and guides, thus promoting a standardized approach.

4. This handbook was prepared by the Services under the sponsorship of the Joint Ordnance Commander's Group (JOCG) E3 Ordnance Safety Subgroup in accordance with the guidelines of the Standardization Reform Policy established by the Secretary of Defense.

5. Comments, recommendations, additions, or deletions and any other pertinent data that may improve this document should be submitted via e-mail marcus.shellman.civ@mail.mil or mailed to:

Defense Information Systems Agency (DISA) Joint Spectrum Center (JSC)
Attn: M. Shellman, Jr., OS35/J5
2004 Turbot Landing
Annapolis, MD 21402-5064

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1. SCOPE

1.1 Purpose. This part of MIL-HDBK-240 includes reference materials for Parts 2 through 7 and presents an introduction to each of these specific parts. MIL-HDBK-240 Parts 1 through 7 supplement MIL-STD-464 by providing guidance for the verification of electromagnetic environmental effects (E3) to ordnance requirements and a methodology for the development of operational guidance related to hazards of electromagnetic radiation to ordnance (HERO). The focus is on E3 ordnance testing for all ordnance and associated support equipment for airborne, sea, space, and ground systems for all Service mission areas. Although intended primarily for use by Department of Defense (DoD) E3 test activities, this handbook also provides a consolidation of "corporate knowledge" about the subject that should be of interest to procurement authorities and system developers. This handbook is only for guidance and cannot be cited as a requirement.

1.2 Background. The specific objectives of this handbook are to:

a. Document E3 Joint-Service test methodology.

b. Promote standardization of ordnance testing and electromagnetic environment (EME) characterization methods.

c. Identify alternative techniques and instrumentation.

d. Facilitate the exchange of E3 test data.

e. Provide methods for developing operational guidance.

1.3 Format. This handbook is issued in parts (see TABLE I). Part 1 gives general information for the required E3 tests to be conducted on ordnance. Parts 2 through 4 provide specific guidance on how to conduct testing to meet the requirements of MIL-STD-464. Part 5 is a placeholder as this content has not yet been developed. Part 6 provides guidance on the characterization of the EME for HERO and Part 7 provides guidance on the documentation of HERO emission control procedures to ensure safety during the conduct of operations and exercises.

TABLE I. MIL-HDBK-240 Parts

PART #	PART TITLE
1	General Guidance
2	Hazards of Electromagnetic Radiation to Ordnance Testing
3	Electrostatic Discharge to Ordnance
4	External Radio Frequency Electromagnetic Environments
5	Lightning Effects to Ordnance *
6	Characterization of the Electromagnetic Environment for HERO
7	Hazards of Electromagnetic Radiation to Ordnance Operational Guidance

* Part 5 had not been developed at the time of publication. NATO Standard NATO Agreement (STANAG) 4370/Allied Environmental Conditions and Test Procedures (AECTP)-504/4 provides assessment guidance for munitions.

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2. APPLICABLE DOCUMENTS

2.1 General. The following documents and publications form a part of this handbook to the extent specified herein. These documents are the most relevant to fully understand the information provided by this handbook. Unless a specific version is referenced, the latest revisions of these documents are to be used. The solicitation or contract should specify the revision of the documents to be used.

2.2 Government Documents.

2.2.1 DoD Specifications, Standards, and Handbooks.

MIL-DTL-23659F	Detail Specification, Initiators, Electric, General Design Specification
MIL-HDBK-235-1	Military Operational Electromagnetic Environment Profiles, General Guidance
MIL-HDBK-235-7	External Electromagnetic Environment Levels for Ordnance
MIL-HDBK-240-2	E3 to Ordnance Guide – Hazards of Electromagnetic Radiation to Ordnance Testing
MIL-HDBK-240-3	E3 to Ordnance Guide – Electrostatic Discharge to Ordnance
MIL-HDBK-240-4	E3 to Ordnance Guide – External Radio Frequency Electromagnetic Environments
MIL-HDBK-240-6	E3 to Ordnance Guide – Characterization of the Electromagnetic Environment for HERO
MIL-HDBK-240-7	E3 to Ordnance Guide – HERO Operational Guidance
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	Electromagnetic Environmental Effects, Requirements for Systems
MIL-STD-882	Standard Practice for System Safety
MIL-STD-1316	Fuze Design, Safety Criteria for
MIL-STD-1377	Effectiveness of Cable, Connector, and Weapon Enclosure Shielding and Filters in Precluding Hazards of Electromagnetic Radiation to Ordnance; Measurement
MIL-STD-1839	Calibration and Measurement Requirements

(Copies of these documents are available online at: https://quicksearch.dla.mil/. For copies of MIL-HDBK-235-7, see Foreword of MIL-HDBK-235-1.)

2.2.2 Other Government Documents and Publications. The following Government publications form a part of this document to the extent specified herein.

(No number assigned.)	Department of Defense Dictionary of Military and Associated Terms
AD 1115	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment
ADS-37A-PRF	Aeronautical Design Standard, E3 Performance and Verification Requirements (Aviation and Troop Command Report)
AFI 91-208	Hazards of Electromagnetic Radiation to Ordnance (HERO)

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	Certification and Management
AR 385-10	The Army Safety Program
DA PAM 385-64	Ammunition and Explosive Safety Standards
DESR 6055.09, Edition 1	Defense Explosives Safety Regulation
DoDI 3222.03	The DoD Electromagnetic Environmental Effects (E3) Program
DoDI 5000.85	Major Capability Acquisition
DoDI 6055.11	Protecting Personnel from Electromagnetic Fields
Joint Requirements Oversight Council Memorandum (JROCM) 102-05, 20 May 2005	Safe Weapons in Joint Warfighting Environments
JOTP-062	Joint Ordnance Test Procedures for PESD and HESD Requirements for Ordnance
NAVSEA OD 30393	Design Principles and Practices for Controlling Hazards of Electromagnetic Radiation to Ordnance
NAVSEA OP 3565/NAVAIR 16-1-529 Volume 1, Sixth Revision	Electromagnetic Radiation Hazards (U) (Hazards to Personnel, Fuel and Other Flammable Material) (U)
NAVSEA OP 3565 Volume 2 Nineteenth Revision	Technical Manual, Electromagnetic Radiation Hazards (U) (Hazards to Ordnance) (U)
NSWCDD ltr 8020, Ser J52/3168, dtd 3 Nov 03	Hazards of Electromagnetic Radiation to Ordnance (HERO) Maximum Allowable Environment (MAE) Analysis Program
STANAG 1380/AECP-2	NATO Naval Radio and Radar Radiation Hazard Manual
STANAG 4370/AECTP-250	Environmental Testing/Electrical and Electromagnetic Environmental Conditions
STANAG 4370/AECTP-500	Environmental Testing/Introduction to Electromagnetic Environmental Verification and Tests
TOP 1-2-511	Electromagnetic Environmental Effects System Testing (Army Test and Evaluation Command)
TR-RD-TE-97-01	EM Effects Criteria and Guidelines for EMRH, EMRO, Lightning Effects, ESD, EMP and EMI Testing of U.S. Army Missile Systems (Redstone Technical Test Center Report)

(Copies of the directives, instructions, and some Joint Publications are available online at: https://assist.dla.mil. If you have any questions, please contact the appropriate ASSIST-Help Desk team: Account/Password Issues: 215-737-8000 [DSN: 442-8000]. Copies of the Defense Explosives Safety Regulation (DESR) are available for download at: https://www.denix.osd.mil/ddes. The JROCM 102-05 can be obtained via direct link to the Navy's E3 Team Online, which is an access restricted, registration approval required site at: https://e3.nswc.navy.mil/e3library/1643895882.pdf. Copies of Naval Sea Systems Command (NAVSEA) documents are available as follows: for DoD agencies and contractors, contact the Director, Naval Ordnance Safety and Security Activity, Attn: N8A, 3817 Strauss Ave., Suite 108, Indian Head, MD 20640-5151 or via https://nossa.dc3n.navy.mil, which is an access restricted, registration approval required site; for all others contact the Commander, Naval Sea Systems Command, Attn: SEA 53H3, 1333 Isaac Hull Avenue, SE, Washington, DC 20376-1080. The direct link for TOP 1-2-511A is available from Defense Technical Information Center

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(DTIC) here: https://apps.dtic.mil/dtic/tr/fulltext/u2/a594352.pdf. Copies of ADS-37A-PRF are available from the U.S. Army CC DEVCOM, Aviation and Missile Center, ATTN: FCDD-AMR-AS-E3, Building 4488, Redstone Arsenal, AL 35898 or via direct link: https://www.avmc.army.mil/Portals/51/Documents/TechData PDF/ADS37APRF.pdf. The direct link for DA PAM 385-64 is available from the Army Publication Directorate website here: https://armypubs.army.mil/epubs/DR_pubs/DR_a/pdf/web/p385_64.pdf. Copies of TR-RD-TE-97-01 are available from the U.S. Army Redstone Technical Test Center, ATTN: TERT-ECE, Building 8978, Redstone Arsenal, AL 35898-8052. Many of the documents also are available at the Defense Information Systems Agency (DISA)/Joint Spectrum Center (JSC) E3 Document Library at this link on the Defense Acquisition University (DAU) website: https://www.dau.edu/cop/e3/Pages/Topics/TopicsMap.aspx.)

2.3 Non-Government Documents.

American National Standards Institute (ANSI)/Institute of Electrical and Electronics Engineers (IEEE)

ANSI/IEEE C63.14	Dictionary of Electromagnetic Compatibility (EMC) including Electromagnetic Environmental Effects (E3)
ANSI/IEEE C95.1	Standard for Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Fields, 3 kHz to 300 GHz
ANSI/IEEE C95.1-2345	Standard for Military Workplaces – Force Health Protection Regarding Personnel Exposure to Electric, Magnetic, and Electromagnetic Fields 0 Hz to 300 GHz
ANSI/IEEE C95.3	Recommended Practice for Measurements and Computations of Radio Frequency Electromagnetic Fields with Respect to Human Exposure to Such Fields, 100 kHz–300 GHz
ANSI/IEEE C95.6	Standard for Safety Levels with Respect to Human Exposure to Electromagnetic Fields, 0 – 3 kHz
IEEE STD 1309-2013	Calibration of Electromagnetic Field Sensors and Probes, Excluding Antennas, from 9 kHz to 40 GHz

(Copies of these documents are available from IEEE on https://www.ieee.org or IEEE Contact Center, 445 Hoes Lane, Piscataway, NJ 08854-1331 or 1-800-678-IEEE.)

International Electrotechnical Commission (IEC)

IEC 61000-4-21	Electromagnetic compatibility (EMC) - Part 4-21: Testing and measurement techniques - Reverberation chamber test methods

(Copies of this document are available from the IEC on https://webstore.iec.ch or +41 22 919 02 11.)

3. DEFINITIONS

3.1 Acronyms and Abbreviations. The acronyms and abbreviations used in this handbook, including all of its parts, are defined below:

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A/m	-	amperes per meter
AC	-	Alternating Current
ADS	-	Aeronautical Design Standard
AECP	-	Allied Environmental Conditions Publication
AECTP	-	Allied Environmental Conditions and Test Procedures
af	-	Audio Frequency
AFB	-	Air Force Base
AFI	-	Air Force Instruction
AFLCMC	-	Air Force Life Cycle Management Center
AIT	-	Automatic Identification Technology
AM	-	Amplitude Modulation
ANSI	-	American National Standards Institute
ASIC	-	Application Specific Integrated Circuits
ASSIST	-	Acquisition Streamlining and Standardization Information System
ATC	-	Army Test Center
ATEC	-	Army Test and Evaluation Command
AUR	-	All-Up Round
AvMC	-	Aviation & Missile Center
BW	-	Bridgewire
CALA	-	Combat Aircraft Loading Area
CAPA	-	Combat Aircraft Parking Area
CCF	-	Cavity Calibration Factor
CDR	-	Critical Design Review
CIWS	-	Close-In Weapon System
CFA	-	Crossed-Field Amplifier
cm^2	-	Centimeter squared
CPCI	-	Constant Peak Conservation Index
CRC	-	Complete Round Code
CS	-	Conducted Susceptibility
CW	-	Continuous Wave
dB	-	Decibel
dBi	-	Decibels with reference to Isotropic Radiator
dBm	-	Decibel-milliwatt
DC	-	Direct Current or Duty Cycle
DD Form	-	Department of Defense Form
DESR	-	Defense Explosive Safety Regulation
DID	-	Data Item Description
DISA	-	Defense Information Systems Agency
DoD	-	Department of Defense
DoDI	-	Department of Defense Instruction

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DoDIC	-	Department of Defense Identification Code
E3	-	Electromagnetic Environmental Effects
E-Field	-	Electric Field
EBW	-	Exploding Bridgewire
EC	-	Equipment Characteristics
EFI	-	Exploding Foil Initiator
EID	-	Electrically Initiated Device
EM	-	Electromagnetic
EMALS	-	Electromagnetic Aircraft Launch System
EMC	-	Electromagnetic Compatibility
EME	-	Electromagnetic Environment
EMF	-	Electromagnetic Frequency
EMI	-	Electromagnetic Interference
EMP	-	Electromagnetic Pulse
EMPAT	-	Electromagnetic Probability-of-Effect Assessment Tool
EMR	-	Electromagnetic Radiation
EMRH	-	Electromagnetic Radiation Hazards
EMRO	-	Electromagnetic Radiation Operational
EMV	-	Electromagnetic Vulnerability
EP	-	Electronic Protection
ERP	-	Effective Radiated Power
ESAD	-	Electronic Safe and Arm Device
ESD	-	Electrostatic Discharge
EW	-	Electronic Warfare
EXCOM	-	Executive Committee
f	-	Frequency
FCC	-	Federal Communications Commission
FCF	-	Firing Consequence Factor
FFAR	-	Folding Fin Aircraft Rocket
FFT	-	Fast Fourier Transform
FI	-	Field Intensity
FM	-	Frequency Modulation
FPGA	-	Field Programmable Gate Arrays
GHz	-	Gigahertz
GPS	-	Global Positioning System
HAZMAT	-	Hazardous Material
HBW	-	Hot Bridgewire
HERF	-	Hazards of Electromagnetic Radiation to Fuel
HERO	-	Hazards of Electromagnetic Radiation to Ordnance
HERP	-	Hazards of Electromagnetic Radiation to Personnel

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:39Z
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HESD	-	Helicopter Electrostatic Discharge
H-Field	-	Magnetic Field
HF	-	High Frequency
HPM	-	High Power Microwave
I/CC	-	Induced/Contact Current
IC	-	Induced Current/Integrated Circuit
IEC	-	International Electrotechnical Commission
IEEE	-	Institute of Electrical and Electronics Engineers
IL	-	Insertion Loss
ISD	-	Ignition Safety Device
JCIDS	-	Joint Capabilities Integration and Development System
JOCG	-	Joint Ordnance Commanders Group
JOERAD	-	Joint Ordnance E3 Risk Assessment Database
J-PRIMES	-	Joint Preflight Integration of Munitions and Electronic Systems
JROCM	-	Joint Requirements Oversight Council Memorandum
JSC	-	Joint Spectrum Center
KA	-	Klystron Amplifier
kHz	-	kilohertz
KMS	-	Knowledge Management System
kV	-	kilovolt
kW	-	kilowatt
LOS	-	Line of Sight
LUF	-	Lowest Usable Frequency
m	-	meter
M&S	-	Modeling and Simulation
mA	-	milliamp
MAE	-	Maximum Allowable Environment
MATDEV	-	Material Developers
MDC	-	Minimum Detectable Current
MECP	-	Multinational Environmental Conditions Publication
MF	-	Multiplying Factor
MHz	-	megahertz
MIL-DTL	-	Military Detail Specification
MIL-HDBK	-	Military Handbook
MIL-STD	-	Military Standard
MK	-	Mark
mm	-	millimeter
MNDC	-	Maximum No-Damage Current
MNF	-	Maximum No-Fire

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MNFC	-	Maximum No-Fire Current
MNFP	-	Maximum No-Fire Power
MNFS	-	Maximum No-Fire Stimulus
MOD	-	Model
MPE	-	Maximum Permissible Exposure
ms	-	millisecond
mW	-	milliwatt
mW/cm^2	-	milliwatts per centimeter squared
NALC	-	Navy Ammunition Logistic Code
NATO	-	North Atlantic Treaty Organization
NAVAIR	-	Naval Air Systems Command
NAVSEA	-	Naval Sea Systems Command
NAWCAD	-	Naval Air Warfare Center, Aircraft Division
NAWCWD	-	Naval Air Warfare Center, Weapons Division
NCF	-	Near-field Correction Factor
NHR	-	No HERO Requirement
NIST	-	National Institute of Standards and Technology
NOSSA	-	Naval Ordnance Safety and Security Activity
ns	-	nanosecond
NSN	-	National Stock Number
NSWCDD	-	Naval Surface Warfare Center, Dahlgren Division
NTIA	-	National Telecommunications and Information Administration
OATS	-	Open Area Test Site
OSH	-	occupational safety and health
P/FM	-	Pass/Fail Margin
P/N	-	Part Number
P-Static	-	Precipitation Static
PCM	-	Pulse-Code Modulation
PDR	-	preliminary design review
PEP	-	Peak Envelope Power
PESD	-	Personnel-borne Electrostatic Discharge
pF	-	picoFarad
PM	-	Pulse Modulation
POE	-	Point of Entry
PPE	-	Personal Protective Equipment
PRF	-	Pulse Repetition Frequency
Q	-	Quality Factor
RADHAZ	-	Radiation Hazard
RAM	-	Radar Absorbing Material
RF	-	Radio Frequency

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RFID	-	Radio Frequency Identification
rms	-	root mean square
RSS	-	root-sum-squares
RTC	-	Redstone Test Center
S&A	-	Safe and Arm
S4	-	Stockpile-to-Safe Separation Sequence
SATCOM	-	Satellite Communications
SCG	-	Security Classification Guide
SE	-	Shielding Effectiveness
SOP	-	Standard Operating Procedure
SRAD	-	Susceptibility RADHAZ Designator
SSD	-	Safe Separation Distance
STANAG	-	Standard NATO Agreement
SUT	-	System Under Test
T/S	-	Transportation/Storage
TOP	-	Test Operations Procedure
TOS	-	Threshold of Susceptibility
TOV	-	Threshold of Vulnerability
TRAD	-	Transmitter RADHAZ Designator
TRR	-	Test Readiness Review
TWT	-	Traveling Wave Tube
TWTA	-	Traveling Wave Tube Amplifier
UHF	-	Ultra High Frequency
UWB	-	Ultra-wide band
UXO	-	Unexploded Ordnance
V/m	-	Volts per meter
VERTREP	-	Vertical Replenishment
VHF	-	Very High Frequency
Vrms	-	Volts root mean square
VSWR	-	Voltage Standing Wave Ratio
VUT	-	Vehicle Under Test
Wi-Fi	-	Wireless Fidelity
WLAN	-	Wireless Local Area Network
WSTC	-	White Sands Test Center
λ	-	Wavelength
μH	-	microHenry
μsec	-	microsecond
Ω	-	Ohm
τ	-	Response Time

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3.2 Definitions. Many of the terms in this handbook are defined in the references listed in paragraph 2. The definitions given below are either not found in the referenced publications or have been included and in some cases expanded from the referenced publication for the convenience of the user, or to make them specific to this handbook.

3.2.1 Acquisition Streamlining and Standardization Information System (ASSIST). The official database containing information about standardization documents used in the DoD. ASSIST also provides electronic access to Government documents included in the database over the Internet. ASSIST can be accessed at https://assist.dla.mil.

3.2.2 All Up Round (AUR). This refers to the completely assembled munition as intended for delivery to a target or configured to accomplish its intended mission. This term is identical to the term all-up-weapon.

3.2.3 Cavity Calibration Factor (CCF). Characteristic value of an enclosed electrically reflective space, which can be used to predict the resultant maximum electric fields (E-fields) as a function of frequency and total radiated power in that space.

3.2.4 Complex Cavity. An electrically large space, where significant reflections of electromagnetic (EM) signals occur from the walls and other structures inside the space; can behave as a reverberant or quasi-reverberant cavity.

3.2.5 Dudding. The inability of the electrically initiated device (EID) to function as intended because the physical/electrical properties have been altered due to the application or repeated application of energy below that required to initiate the device.

3.2.6 Electrically Initiated Device (EID). A single unit, device, or subassembly that uses electrical energy to produce an explosive, pyrotechnic, thermal, or mechanical output. Examples include electroexplosive devices such as hot bridgewire (HBW), semiconductor bridge, carbon bridge, and conductive composition, as well as exploding foil initiators (EFIs), laser initiators, burn wires, and fusible links.

3.2.7 Electromagnetic Environment (EME). The environment created by man-made sources (electrical and electronic systems, radio frequency (RF) systems, radar and high-power microwave (HPM) systems, and so forth) and by natural sources (lightning, precipitation static (p-static), electrostatic discharge (ESD), galactic and stellar noise, and so forth) encountered by a military force, system, or platform when performing its assigned mission in its intended operational environment. In the case of ordnance, this is during its stockpile-to-safe-separation sequence (S4), as defined in 3.2.35. Specific equipment characteristics (that is, operating frequencies and power levels) and operational factors (that is, distances between items (boxes, platforms, systems, and so forth) and frequency coordination) all contribute to this environment.

3.2.8 Electromagnetic Environmental Effects (E3). E3 is the impact of the EME upon the operational capability of military forces, equipment, systems, and platform/systems. It encompasses all electromagnetic disciplines, including electromagnetic compatibility (EMC); electromagnetic interference (EMI); electromagnetic vulnerability (EMV); electromagnetic pulse (EMP); electronic protection (EP); electrostatic discharge (ESD); and hazards of electromagnetic

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radiation to personnel (HERP), ordnance (HERO), and volatile materials such as fuel (HERF); and includes the electromagnetic effects generated by all EME contributors including radio frequency (RF) systems; ultra-wideband (UWB) devices; high-power microwaves (HPM) systems; lightning; precipitation static (p-static); and so forth.

3.2.9 Electrostatic Discharge (ESD). A transfer of electric charge between bodies of different electrostatic potential in proximity or through direct contact.

3.2.10 Field Calibration. The process of determining average and peak EME levels for testing using required transmitter modulation, power, and frequency with aligned antenna polarization.

3.2.11 Far-Field Region. The region of the field of an antenna where the angular field distribution is essentially independent of the distance from the antenna. The electric and magnetic field (H-field) components are mutually perpendicular to each other and the direction of propagation, and their ratio is 377 ohms (Ω), the impedance of free space. The far-field distance generally is recognized as equal to $2D^2/\lambda$, where D is the largest dimension of the antenna and λ is the wavelength of the radiating energy.

3.2.12 Hazards of Electromagnetic Radiation to Ordnance (HERO). The situations in which exposure of ordnance to external EMEs results in specified safety or reliability margins of EIDs or electrically powered ordnance firing circuits to be exceeded, or EIDs to be inadvertently actuated. External EMEs may originate from intentional transmitting sources (for example, radios, radars, electronic countermeasures equipment) or unintentional sources (for example, arcing, high current switching transients). Consequences include both safety (premature firing) and reliability (EID dudding or altered functional characteristics) effects.

3.2.13 Helicopter-borne Electrostatic Discharge (HESD). The sudden flow of electric charge between a helicopter or rotary winged aircraft and an object of different electrical potential. A buildup of static electricity can be caused by triboelectric charging or electrostatic induction generated from operating rotary wings.

3.2.14 HERO Margin. The difference between the maximum no-fire stimulus (MNFS) and the permissible EID response level. For EIDs with a safety consequence, the margin is defined in MIL-STD-464 as 16.5 decibels (dB); for EIDs with a reliability consequence, the margin is defined as 6 dB.

3.2.15 HERO SAFE ORDNANCE. Any ordnance item that is proven by test or analysis to be sufficiently shielded or otherwise so protected that all EIDs contained by the item are immune to adverse effects when the item is employed in the EME delineated in MIL-STD-464.

3.2.16 HERO Safe Separation Distance (SSD). The minimum distance between the ordnance and transmitting antenna, beyond which the EME is less than or equal to the ordnance's maximum allowable environment (MAE).

3.2.17 HERO SUSCEPTIBLE ORDNANCE. Any ordnance item containing EIDs proven by test or analysis to be adversely affected by radio frequency energy to the point that the

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safety or reliability of the system is in jeopardy when the system is employed in the EME delineated in MIL-STD-464.

3.2.18 HERO UNSAFE ORDNANCE. Any ordnance item containing EIDs that have not been classified as HERO SAFE or HERO SUSCEPTIBLE as a result of a HERO analysis or test, to include lack of testing. Additionally, any ordnance item containing EIDs (including those previously classified as HERO SAFE or HERO SUSCEPTIBLE ordnance) that has its internal wiring exposed; when tests are being conducted on that item that result in additional electrical connections to the item; when EIDs having exposed wire leads are present and handled or loaded in any but the tested condition; when the item is being assembled or disassembled; or when such ordnance items are damaged causing exposure of internal wiring or components or destroying engineered HERO protective devices.

3.2.19 Launcher. Any structure or system as part of a weapon or weapon system particular to an ordnance item from which it can be fired, launched, or deployed to initiate that ordnance item's intended purpose.

3.2.20 Maximum Allowable Environment (MAE). The highest radiated field strength levels to which ordnance can be exposed without reducing HERO margins.

3.2.21 Maximum No-Damage Current (MNDC). The maximum no-fire stimulus (MNFS) applicable to exploding foil initiators (EFIs) that will not result in visible damage or cause an EFI not to fire when subjected to a firing pulse of more than 0.1% of all EFIs of a given design at a confidence level of 95%.

3.2.22 Maximum No-Fire Current (MNFC). The MNFS applicable to EIDs whose normal performance is specified in terms of current.

3.2.23 Maximum No-Fire Stimulus (MNFS). The greatest firing stimulus that will not cause initiation or degrade an EID of more than 0.1 percent of all electric initiators of a given design at a confidence level of 95 percent. Stimulus refers to electrical parameters such as current, rate of change of current (di/dt), power, voltage, or energy, which are most critical in defining the no-fire performance of the EID.

3.2.24 Near-Field Region. The reactive region immediately surrounding the antenna, wherein the reactive field dominates, and a radiating region where the angular field distribution is dependent on the distance from the antenna. The near-field region is typically more complex than the far-field region. In the near field, coupling to an ordnance item is expected to be influenced by the type of radiating antenna and the proximity to that antenna. If the antenna has a maximum overall dimension that is not large compared to wavelength, then this region may not exist.

3.2.25 No HERO Requirement (NHR). Ordnance that does not contain EIDs.

3.2.26 Ordnance. Explosives, chemicals, pyrotechnics, and similar stores (for example, bombs, guns, ammunitions, flares, smoke, and napalm) carried on an airborne, sea, space, or ground system.

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3.2.27 Ordnance Configurations. The physical configurations assumed by the ordnance and its host platform/systems (for example, aircraft, ground vehicles, personnel, and so forth) and associated ancillary equipment throughout the operational S4. Examples include the ordnance item in its shipping/storage container, the item loaded onto its launch platform, or a bare missile in a post-launch scenario.

3.2.28 Painted Illumination. The movement of the test antenna with the transmitter constantly on and radiating such that the entire surface area of the system under test (SUT), or the portion of the SUT under test, is illuminated. Also known as painting.

3.2.29 Personnel-borne Electrostatic Discharge (PESD). The sudden flow of electric charge between personnel and an object of different electrical potential. A buildup of static electricity can be caused by triboelectric charging or electrostatic induction generated by the movement of the person's body.

3.2.30 Platform. A mobile or fixed installation such as a ship, aircraft, ground vehicles and shelters, personnel, launch-space vehicles, shore or ground station. For the purposes of this document, platforms are limited to those platforms on which a weapon or weapon system can be mounted, transported, or employed.

3.2.31 Reliability Consequence. The inadvertent actuation of an EID that does not result in a safety consequence, but degrades system performance; that is, renders the ordnance item either ineffective or unable to function as intended. An example of this would be the RF initiation of a detonator in a fuze whose safe and arm (S&A) device is mechanically out-of-line with the explosive train. Another example of an EID with a reliability consequence is an electrically initiated match in a thermal battery which, when activated, simply initiates the chemical process to stimulate the battery. In addition, the definition has been expanded to include dudding where the system would no longer be reliable.

3.2.32 Rise Time. The time required for the EID bridgewire temperature to increase from 10 to 90 percent of its final value in response to a step-function input current.

3.2.33 Safety Consequence. The inadvertent actuation of an EID that creates an immediate catastrophic event with the potential to either destroy equipment or injure personnel, such as the firing of an in-line rocket motor igniter by RF energy or the inadvertent actuation of an EID that increases the probability of a future catastrophic event by removing or otherwise disabling a safety feature of the ordnance item. This, for example, might be caused by the RF initiation of a piston actuator that removes a lock on the S&A rotor of an artillery fuze, thus allowing a sensitive detonator to rotate in-line with the explosive train.

3.2.34 Safety-Critical Function. A function whose failure to operate or incorrect operation will directly result in a mishap of either Catastrophic or Critical severity.

3.2.35 Stockpile-to-Safe Separation Sequence (S4). The progressive stages (phases) that begin at the time the ordnance is manufactured and continue until it is expended or reaches a safe distance from the launch vehicle/platform/system (see FIGURE 1).

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FIGURE 1. Stockpile-to-safe separation sequence (S4)

This progression is referred to as the S4 and may consist of up to six of the following distinct stages in which varying degrees of susceptibility can result from unique physical configurations or operational EMEs:

3.2.35.1 Transportation/Storage. The phase in which the ordnance is packaged, containerized, or otherwise prepared for shipping or stored in an authorized storage facility. This includes transporting of the ordnance.

3.2.35.2 Assembly/Disassembly. The phase involving all operations required for ordnance build-up or breakdown and typically involves personnel.

3.2.35.3 Staged. The phase where the ordnance has been prepared for loading and is pre-positioned in a designated staging area.

3.2.35.4 Handling/Loading. The phase where physical contact is made between the ordnance item and personnel, other electronics, metal objects or structures during the process of preparing, checking out, performing built-in tests, programming/reprogramming, installing, or attaching the ordnance item to its end-use platform/system; for example, aircraft, launcher, launch vehicle, or personnel. These procedures may involve making or breaking electrical connections, opening and closing access panels, removing/installing safety pins, shorting plugs, clips, and dust covers. This configuration also includes all operations required for unloading; that is, removing, disengaging, or repackaging the ordnance item.

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3.2.35.5 Platform Loaded. The phase where the ordnance item has been installed on or attached to the host platform/system, (for example, aircraft, ground vehicles, personnel, and so forth) and all loading procedures have been completed.

3.2.35.6 Immediate Post-Launch. The phase where the ordnance item has been launched from its platform/system but has not reached its safe separation distance with regard to the actuation of its explosives, pyrotechnics, or propellants.

3.2.36 System Under Test (SUT) Operating Frequencies. Frequencies inherent to the operation of the ordnance SUT, such as computer clock frequencies, internal oscillator frequencies, and communication frequencies.

3.2.37 Thermal Stacking. Heating of the bridgewire followed by a relaxation period where some cooling occurs. After several thermal time constants, the temperature of the EID bridgewire reaches an equilibrium condition with some small temperature excursions about the equilibrium point.

3.2.38 Weapon System. An ordnance item including the components necessary for its proper function in deploying, launching, or firing. A weapon system can include a single weapon or multiple weapons, platform(s), launcher(s) and the appropriate ordnance for each weapon, for example, M1A2 Abrams main battle tank, H-60 series helicopters, or the F-35 Joint Strike Fighter (JSF) aircraft.

4. JOINT ORDNANCE COMMANDERS GROUP (JOCG) E3 ORDNANCE SAFETY SUBGROUP

4.1 Purpose. The JOCG provides a forum where all conventional ammunition life-cycle stakeholders, including the top process owners from its member commands and associated or related organizations, jointly define and improve munitions systems. The underlying objective is to identify, implement or recommend for implementation, joint opportunities to reduce cost, increase effectiveness and ensure interoperability and/or interchangeability of munitions systems. The JOCG E3 Ordnance Safety Subgroup was established under the JOCG Executive Committee (EXCOM) to develop and maintain technical safety information, tools, data sharing, and standardized test methodologies for the development, testing, and acquisition and deployment of ordnance in an electromagnetic environment. Additionally, it is intended to influence harmonization and standardization with respect to the dissemination of safety information and additional guidance across the DoD.

4.2 Background. The JOCG E3 Ordnance Safety Subgroup was established in 2014 and replaced the JOCG HERO Subgroup that was established in 1994. Previous work of the JOCG HERO Subgroup included the development of MIL-HDBK-240, the Joint Ordnance E3 Risk Assessment Database (JOERAD) database, the development of the MAE Tool, and the Joint HERO curves, all of which were HERO focused efforts. The transformation of the JOCG HERO Subgroup into the JOCG E3 Ordnance Safety Subgroup was to ensure that in addition to HERO, ESD, EMV, and Lightning were addressed. This led to the growth of MIL-HDBK-240 to include all of these E3 Ordnance Safety disciplines. In addition, the JOCG E3 Ordnance Safety Subgroup has stood up a Service review process for HERO testing such that all DoD HERO test

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reports are reviewed and a Joint HERO Classification (that is, HERO SAFE, HERO SUSCEPTIBLE, HERO UNSAFE) is provided to ensure harmonization/commonality for the provision of operational guidance in a Joint environment. The results of this review also ensure that all HERO data provided to the Joint ordnance database has been reviewed and endorsed across the Services. Similarly, the testing and intended use of RF emitters for inventory tracking of ordnance are reviewed and subsequent HERO SSDs are determined; the Subgroup then provides concurrence and recommends certification for the use of these devices throughout the DoD. The result of each of these efforts is a letter issued on behalf of the E3 Ordnance Safety Subgroup by the Chairman of the Subgroup. Similar to HERO, the JOCG's future efforts will include a review of the Services' ESD testing to ensure harmonization/ commonality and to ensure that all of the ESD data is provided to the Joint database.

4.3 Mission Statement. The mission of the JOCG E3 Ordnance Safety Subgroup is to provide the DoD and its Allies a common understanding of the operational EME, a common approach for material acquisition E3 ordnance testing, and a Joint implementation approach to properly and consistently manage E3 to ordnance and weapons systems safety and interoperability during the conduct of operations and exercises.

The JOCG E3 Ordnance Safety Subgroup's top priority will be to develop and maintain technical safety information, tools, data sharing, and standardized test methodologies for the development, testing, and acquisition and deployment of ordnance. Additionally, it will influence harmonization and standardization with respect to the dissemination of safety information and additional guidance across the DoD.

4.4 E3 Activities. Table II identifies key Army, Navy, and Air Force activities, along with their respective responsibilities in each Service's E3 Ordnance Safety programs. (See MIL-HDBK-237 for additional guidance.) The operational EMEs are defined in MIL-STD-464 and those environments are rolled-up from the detailed EMEs defined in MIL-HDBK-235. Many of the Service environments defined within MIL-HDBK-235, particularly the naval environments, are a result of instrumented measurements and related efforts as described in Part 6. The combined efforts described in Parts 2 and 6 then allow for the provision of operational guidance as described in Part 7. The ESD test environment is also defined in MIL-STD-464.

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TABLE II. DoD Organizations Concerned with E3 Ordnance Safety

Service	Facility	Location	Responsibilities
Army	Combat Capabilities Development Command (CCDC), Aviation & Missile Center (AvMC)	Redstone Arsenal, AL	Aircraft
	HQ, Army Test and Evaluation (ATEC) Command	Aberdeen Proving Ground, MD	All Army Ordnance Systems
	ATEC, Redstone Test Center	Redstone Arsenal, AL	Missiles, ground vehicles, and Army aircraft
	CCDC, Armaments Center	Picatinny Arsenal, NJ	Fuzes, gun ammunition, mines, simulators, and infantry ordnance
	ATEC, White Sands Test Center	White Sands, NM	Missiles, ground vehicles, and soldier ordnance
Navy & U.S. Marine Corps	Marine Corps Systems Command (MARCORSYSCOM)	Quantico, VA	USMC Defense Systems Acquisition Command
	Naval Surface Warfare Center, Dahlgren Division	Dahlgren, VA	Assessments of ordnance and HERO surveys and operational guidance
	Naval Ordnance Safety and Security Activity (NOSSA)	Indian Head, MD	DON HERO Program Manager, HERO Technical Warrant Holder and Certification authority
	NAVAIR Patuxent River	Patuxent River, MD	Aircraft E3
	Naval Surface Warfare Center, Indian Head Division	Indian Head, MD	DON ESD Technical Agent
Air Force	Air Force SEEK EAGLE Office (AFSEO)	Eglin AFB, FL	Aircraft and ordnance compatibility
	Air Force Safety Center (HQ AFSEC/SEWN)	Kirtland AFB, NM	Nuclear Weapons Branch
	Warner Robins Air Force Base (AFB)	Warner Robins AFB, GA	Rotary-wing aircraft
	Wright-Patterson AFB	Wright-Patterson AFB, OH	Fixed-wing aircraft
	85th Engineering Installation Squadron	Kessler AFB, MS	Electromagnetic Surveys

5. HERO OVERVIEW – PART 2

5.1 The Need for HERO Control. Technological advances have resulted in the proliferation of communication-electronic equipment and the development of higher power

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equipment that radiate EM energy. These advances, coupled with the trend to utilize more sensitive, low-power electronic circuits in the design of ordnance systems, perpetuate a long-standing hazard. The hazards that result from adverse interactions between the EME and the electrical initiators or initiating systems contained within ordnance systems are referred to in DoD terminology as HERO. The HERO problem arises from a fundamental incompatibility between the EIDs or EID firing circuits contained within the ordnance and the external radiated EME that the ordnance encounters during its S4 progression.

EIDs perform a variety of functions, such as initiating rocket motors, arming and detonating warheads, and ejecting chaff and flares. The HERO problem arises when any of these functions occur unintentionally or prematurely, as a result of exposure to EM energy. There are two potential forms of such unintentional, RF-induced EID response:

a. Activation of the initiating device itself by EM energy coupled directly into the device or upset of an energized firing circuit, resulting in a firing signal erroneously being sent to the EID.

b. Degradation or dudding of the initiating device by EM energy coupled directly into the device.

In case (a), accidental EID activation can have negative consequences on either safety or performance. A safety consequence is the inadvertent actuation of an EID that creates an immediate catastrophic event that has the potential to either destroy equipment or injure personnel, such as the firing of an inline rocket motor igniter; or the inadvertent actuation of an EID that increases the probability of a future catastrophic event by removing or otherwise disabling a safety feature of the ordnance item. This, for example, might be caused by the RF initiation of a piston actuator that removes a lock on the S&A rotor of an artillery fuze, thus allowing a sensitive detonator to rotate in-line with the explosive train.

Performance degradation (b) can be any condition that does not have safety implications and is also referred to as “reliability.” Performance degradation may occur because an EID has been desensitized as a result of multiple low-level exposures, which would prevent it from firing when needed, or because it already had been ignited. Even if the EID has been ignited, this could still be considered as performance degradation if, for example, the EID ignites a thermal battery, which would cause the ordnance to not operate properly but is not considered a safety issue.

The combination of severe EME levels and sensitive, insufficiently protected components/ circuits can have disastrous consequences. Although the problem was recognized in the late 1950s, it has persisted even today for three reasons:

a. Proliferation of communication-electronic equipment.

b. Introduction of more powerful emitters, raising operational EME levels.

c. Use of sensitive electrically initiated systems.

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Today, MIL-STD-464 requires that ordnance be both designed with sufficient protection from the EME and that performance be verified by test or analysis. "Safety" and "reliability" EID categorizations should be determined by the procuring activity.

5.2 Approaches to Addressing the HERO Problem. Because of the varied experiences with HERO within the Services, it is not surprising that Service-unique approaches have evolved over time to deal with the problem. Besides Service history, other factors have influenced the respective HERO programs, such as the way the Services store, transport, and use ordnance, and the practical options available for minimizing hazards. For example, when operational EME levels exceed susceptibility thresholds, the Services can opt to utilize different risk reduction measures. The Army and Air Force might stipulate a minimum separation distance between the susceptible ordnance and the offending transmitter, whereas, limited space aboard naval platform/systems might leave no other option for the Navy than to impose restrictions on the emissions of the offending transmitter(s); for example, reducing transmitter output power or limiting antenna radiation zones.

5.2.1 Commonality. Despite differences in the way each Service manages HERO, there are certain essential elements that are common to all Services' HERO programs. These include the following:

a. A definition of the expected EME levels for all ordnance configurations.

b. Prescribed methods to quantify system degradation (deficiencies).

c. A process to develop and validate effective, practical "fixes" for known EM deficiencies.

Part 2 of this handbook focuses primarily on (b) above. The other elements are addressed in one or more of the various documents listed: MIL-STD-464, AD 1115, ADS-37A-PRF, NAVSEA OD 30393, NAVSEA OP-3565, AFI 91-208, and DA PAM 385-64.

The HERO EME levels for design and performance verification are found in MIL-STD-464. However, each Service uses somewhat different processes to address HERO. This handbook prescribes recommended methodology and procedures that should be adopted by all test activities concerned with the planning and execution of HERO tests.

5.2.2 Program Manager Responsibilities. DoDI 3222.03 states that HERO will be mitigated before the administration of military exercises, operations, and activities and this includes HERO certification testing of ordnance, the characterization of the operational EME and the provision of operational guidance. Thus, there are a number of activities within each Service that are assigned various HERO program responsibilities. These responsibilities include both administrative and technical aspects of the program. Activities with assigned technical roles generally are concerned with determining whether the system design is adequate to satisfy the E3 requirements of MIL-STD-464. This most often is accomplished by Government-controlled tests at Government-operated test facilities as described in Part 2. The purpose of certification is to ensure that the ordnance will operate safely within the EME levels anticipated during the service life of the ordnance, and that any susceptibilities are quantified and documented.

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5.3 HERO Certification/Verification Testing. The purpose of a HERO test is to determine if exposure of EIDs to specified EME levels will affect the ordnance in an adverse way. As mentioned earlier, there are several types of adverse effects. Unintentional activation of EIDs within the ordnance can result in a hazard or degrade reliability. It is possible that such adverse effects can occur from exposure to EME levels, whether or not power is applied to the firing circuits. The Army historically distinguished the non-energized problem by the acronym Electromagnetic Radiation Hazards (EMRH). On the other hand, if power is applied, firing circuits may experience interference to an extent that erroneous firing commands are issued to the EIDs, historically referred to by the Army as Electromagnetic Radiation Operational (EMRO). There is also an EMROH (old term) environment that denotes the time just prior to launch where firing circuits are active; at this point, the system EME has a 6 dB margin placed on it for safety. *It is noted that the term HERO will be used henceforth in this handbook in a comprehensive sense to represent both cases; that is, firing circuits powered or unpowered.* Any distinctions between these two cases with respect to test methods, ordnance configurations, ordnance procedures, susceptibility thresholds, and so forth, will be made clear in the context of the discussion.

The general approach for HERO testing is to expose inert, instrumented ordnance to a controlled test EME and monitor each EID contained within the ordnance for a possible response. In addition, when active (powered-up) firing circuit testing is necessary, the firing circuit should be monitored. Upset to the firing circuit is typically go/no-go in nature. For most EIDs, the response is quantified in terms of the magnitude of RF current induced into the heating element, or bridgewire, of the device. Measured current levels are determined as a function of selected environment parameters and, in some cases, specific test procedures used to exercise the ordnance. A common objective in all HERO testing is to determine the maximum or worst-case response at each test frequency for various ordnance physical configurations. The general approach is to establish a desired test EME level at a selected test frequency and record the EID response as each of several test parameters; for example, illumination angle, polarization, and so forth, are varied. Specific test procedures may vary according to the type of test facility being used; for example, open-air site, reverberation chamber, or anechoic chamber. Part 2 of this handbook provides details concerning the steps necessary to plan a test, instrument the ordnance item, generate the test environment, conduct the test, and analyze the test results.

HERO testing should emphasize exposure of the ordnance to the EME levels that are associated with each phase of an ordnance item's S4, as defined in 3.2.24. FIGURE 1 illustrates a typical progression through this sequence. Significant differences in the ordnance item's physical configuration can be expected as the item transitions from one phase to another.

Different physical configurations can be expected to offer different levels of protection. Furthermore, it is likely that the EME associated with each phase will be quite different. For example, the EME levels associated with handling/loading operations are generally less than those encountered during certain other phases, such as transportation/storage. Thus, the potential for a HERO problem is highly dependent on both of these phase-dependent conditions. From a HERO test standpoint, it is especially important to test all unique ordnance configurations.

5.4 Data Repositories. HERO data is documented in activity test reports, which typically are provided to the individual program office that funded the test. There are some

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notable differences in the way each Service maintains and disseminates HERO data. In the case of the Navy, HERO reports initially are provided to the Naval Ordnance Safety and Security Activity (NOSSA). NOSSA reviews the report and, if approved and certified, forwards it to the cognizant program office. The Army uses the U.S. Army Test and Evaluation Command (ATEC) to issue safety releases and the Combat Capabilities Development Command (CCDC) Aviation & Missile Center (AvMC) issues E3 Airworthiness releases. During the acquisition process, Army Materiel Developers (MATDEVs) are responsible for obtaining HERO evaluations and certifications of military munitions both for routine mission profiles and for anticipated Joint or combined operational missions. The Air Force typically incorporates safety-critical HERO information into individual weapon Technical Manuals, whereas the Navy and Marine Corps consolidate all HERO data to the Fleet via the E3 Team Online Knowledge Management System (KMS). HERO test results also are reviewed by the Navy's Weapon System Explosive Safety Review Board as a prerequisite for release for Service use. As a relatively new process, all Services are now required to provide HERO data to the DISA/JSC for entry into the joint HERO data repository.

6. ESD OVERVIEW – PART 3

6.1 Program Manager Responsibilities. DoDI 3222.03 and MIL-STD-464 require that ordnance not be inadvertently initiated or dudded by electrostatic discharge. ESD (for the purposes of this handbook) is divided into two types, PESD and HESD, and the specific environments for each are defined in MIL-STD-464. Part 3 of this handbook describes the PESD and HESD test procedures for all ordnance/munition (non- nuclear) types for the purpose of Joint Service assessment. These procedures also specify the PESD and HESD requirements for all ordnance items including bare energetic materials that are cataloged and/or fielded with locally assigned stock numbers or national stock numbers (NSNs).

6.2 PESD. The purpose of PESD testing is to determine if exposure of ordnance items, ordnance electrical interfaces, and related systems, including but not limited to firing circuits and fire control electronics to the PESD test requirements results in a safety and/or operability hazard. These hazards may be ordnance or EID initiation, firing circuit initiation, performance degradation, electrical interface malfunction, or dudding according to MIL-STD-464 and MIL-STD-331. PESD exposure is assessed or tested for bare energetic materials, bare devices, and bare AURs in the worst-case operational mode during any of the S4 phases (that is, transportation/storage, assembly/disassembly, staged, handling/loading, and platform-loaded).

6.2.1 Application. Data obtained through the use of this document should be included in the ordnance item/munition safety data package. The PESD test requirements are applicable to all ordnance items (with and without EIDs), ordnance subsystems, ordnance-related equipment, launchers, remote control devices, EIDs and firing/monitoring circuits, including those items installed external to aircraft and vehicles.

PESD test requirements may be applicable at the equipment /system/AUR level and/or the device level depending on the configuration, assembly and use of the item. The PESD requirements are applied to selected test points on the test item which, without regard to procedural controls for the test item (such as a requirement to wear a static-dissipative wrist strap and ground before handling the item), could possibly be contacted by personnel during the

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different configurations/modes of operation including: transportation/storage, assembly/disassembly, staged, handling/loading, and platform-loaded. Unauthorized non-standard configurations do not need to be addressed such as disassembly of an AUR that is not authorized in the field.

6.2.2 PESD Limitations. This document is only applicable to conventional (non-nuclear) munitions to aid in the item, device, system, or AUR safety and operability assessment. This document is not intended to be used to aid the assessment of effectiveness, reliability or performance of an ordnance item/AUR.

6.3 HESD. The purpose of HESD testing is to determine if exposure of ordnance items, ordnance electrical interfaces, and related systems, including but not limited to firing circuits and fire control electronics (weapon release, intent to launch, and so forth) results in a safety and/or operability hazard such as ordnance or EID initiation, firing circuit initiation, performance degradation, electrical interface malfunction, or dudding due to an operating helicopter and/or aircraft exposure in accordance with the requirements of MIL-STD-464 and MIL-STD-331.

HESD generally refers to one or more of the following configurations: vertical replenishment (VERTREP) (packaged/shipping/storage), bare/man-carry (out of shipping container and carried into or out of operating aircraft), external-carry/captive-carry (mechanically and electrically attached to aircraft and exposed to in-flight charging), and hot-tube-loaded (mechanically or electrically loaded or unloaded to an operating ungrounded aircraft).

6.3.1 Application. Data obtained through use of this document is included in the ordnance item/munition safety data package. The HESD test procedures are applicable to all ordnance items (with and without EIDs), ordnance subsystems, ordnance-related equipment, launchers, remote control devices, EIDs and firing/monitoring circuits, including those installed external to aircraft and vehicles, those carried outside the shipping container internal to the aircraft, and for hot-tube-loading operations.

HESD test procedures may be applicable at either or both the equipment/system/AUR level and the device level depending on the configuration, assembly, and use of the item in the field. The HESD requirements are applied to selected test points on the test item without regard to procedural controls for the test item (such as a requirement to wear a static-dissipative wrist strap and ground before handling the item), which could possibly make contact with a charged aircraft or become charged during the different configurations/modes of operation including: transportation/VERTREP/sling-load, man-carry transporting (internal, outside the shipping container), external carry, and hot-tube-loading/unloading. Unauthorized non-standard configurations do not need to be addressed such as disassembly of an AUR that is not authorized in the field. Specific HESD test requirements are identified by ordnance category.

6.3.2 HESD Limitations. This document is only applicable to conventional (non-nuclear) munitions to aid in the item, device, system, or AUR safety and operability assessment. This document is not intended to be used to aid the assessment of effectiveness, reliability or performance of an ordnance item/AUR.

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7. EXTERNAL RF EME OVERVIEW – PART 4

7.1 Program Manager Responsibilities. DoDI 3222.03 and MIL-STD-464 require that ordnance meet its operational performance requirements without experiencing unacceptable performance degradation as a result of exposure to the external RF EME. Thus, there are a number of activities within each Service that are assigned various responsibilities relative to the assessment of ordnance performance in the external RF EME. These responsibilities include both administrative and technical aspects of the program. Activities with assigned technical roles generally are concerned with determining whether system design is adequate to satisfy the external RF EME requirements of MIL-STD-464. This most often is accomplished by Government-controlled tests at Government-operated test facilities as described in Part 4 where ordnance is evaluated in the full threat environment; however, other evaluation methods such as the use of analysis, modeling and simulation (M&S), and/or supplemental test techniques involving use of subsystem-level testing and electromagnetic shielding effectiveness (SE) may be employed.

7.2 External RF EME Verification. The purpose of external RF EME evaluation is to demonstrate that ordnance will not be adversely affected when exposed to its operational EME during the employment of the system. The performance of the system can be influenced by three principle factors: (1) EME characteristics, (2) physical configuration(s) of the ordnance, and (3) operational electronics. Each factor should be considered from the time the ordnance is fielded until it completes its life cycle when assessing performance in the external RF EME. Furthermore, when the design of the ordnance and/or platform is altered, it is critical that a detailed design review be conducted to properly assess the impact of the changes.

7.3 Verification Approaches/Procedures. The typical approach for verifying system performance is to conduct testing with an inert test asset during which all critical electronic functionality of the ordnance is exercised in the presence of a controlled test EME and system performance is monitored via the use of specialized telemetry equipment and/or real-time assessment. Since system responses are highly dependent on the system test configuration, significant effort should be put forth in configuring test assets and test setups, to include the interface between the ordnance and any launcher and/or launch platform, in order to obtain data that is an accurate reflection of system performance. While testing is generally recognized as the preferred means of determining system performance, testing can have limitations, which may include a practical limit of the number of test frequencies and/or hardware configurations that can be evaluated within the timeframe of the test event, and the inability of the test lab to generate the full external RF EME. Therefore, other evaluation methods such as analysis, M&S, and/or supplemental test techniques involving the use of subsystem-level testing and electromagnetic SE may be required to supplement data obtained from external RF EME testing. Additional discussion of these approaches for evaluating system performance in the external RF EME is provided in Part 4.

8. LIGHTNING – PART 5 (CONTENT NOT DEVELOPED AT THIS TIME)

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9. EVALUATION OF THE OPERATIONAL EME OVERVIEW – PART 6

9.1 Introduction. As detailed in Part 6, the characterization of the operational EME is possible through either instrumented measurements or calculations using the specific transmit parameters of the transmitter/antenna combination, or a combination of both measurements and calculations. Usually, instrumented measurements provide a more accurate characterization of the EME and may result in less restrictive operational guidance. These assessments characterize the level of RF hazards and provide guidance to minimize these hazards while maintaining a level of safety in the RF environment. Inasmuch as many items have some level of susceptibility to the operational EME and/or are classified as HERO SUSCEPTIBLE, the program requirements do not stop at the conclusion of the HERO test. Since the operational EME is always changing, HERO is not a "test and forget" program and as items are routinely exposed to EMEs that exceed ordnance susceptibility thresholds, further efforts are required to ensure safety and address risk during the conduct of operations and exercises. These include the characterization of the operational EME and finally the provision of operational guidance (that is, operational procedures, restrictions, frequency and/or power management to control the EME, or SSDs, reference Part 7 of this handbook).

For HERO, actual measurements of the EME are typically conducted during a "HERO Survey" of a ship or installation or activity. Due to the proliferation of wireless systems currently being used, measurements of the RF environment may also include measurements in enclosed reflective spaces to characterize the complex EMEs produced inside these cavities of a reverberant nature. Examples of these cavities are transportation containers (including semi-rig trailers) and ship storage rooms. It is critical to assess the resultant EME in confined, highly reflective, complex cavities, especially where potential hazards to ordnance may exist. Transportation vehicles are commonly tracked now using Global Positioning Systems (GPSs), with radio frequency identification (RFID) tag systems attached to the ordnance items. In all instances, the choice of measurement equipment and measurement techniques is critical and should be carefully considered prior to the conduct of measurements.

In addition to the operational guidance generated from the HERO test results and characterization of the operational EME, the results from the characterization of the EME are used to update MIL-HDBK-235 and the EME tables found in MIL-STD-464 as these environments drive the HERO test environment. Additionally, the data is used to update JOERAD and are contributing to North Atlantic Treaty Organization's (NATO's) efforts to capture the NATO operational EME.

9.2 HERO Surveys. HERO surveys are more than just measurements of the EME, providing the complete HERO posture of the platform or facility by identifying all of the ordnance containing EIDs and associated platforms; the locations where ordnance is stored, transported, and used; specific mission requirements; and locations for all of the transmit antennas. HERO surveys provide a realistic real-world basis for guidance with regard to EME controls for ordnance, whereas theoretical assessments/calculations tend to be more restrictive.

Surveys are comprised of in-briefs, tours of the ship or installation, documentation of antenna/transmitter and ordnance operations and locations, measurements and calculations of the EME, and exit briefs at the conclusion of the survey period.

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The overall approach to conduct HERO surveys is a five-step process. This process begins with test planning and is followed by a progression of activities to execute the test; perform a test assessment; analyze, reduce, and evaluate the data; and report the results. The specifics of these are presented in Part 6.

10. HERO OPERATIONAL GUIDANCE OVERVIEW – PART 7

10.1 Introduction**.** Using both the MAE data for ordnance and measured and/or calculated EMEs at the ordnance locations, HERO can be managed either through emission control procedures (that is, frequency and/or power management or sectoring, procedural controls or other restrictions) or through calculated SSDs. The latter option of calculating EMEs and providing SSDs is purely analytical and is generally a more conservative approach for managing HERO (particularly in a shipboard environment), and the result is generally more restrictions and less operational flexibility. Historically, HERO surveys have provided the added benefit of measuring the actual EME that exists at ordnance locations, providing a more exact means for mitigating the HERO issue and precise guidance for specific ordnance evolutions. That said, in the shipboard environment, SSDs are reserved for mitigating HERO for in-flight aircraft carrying ordnance and for managing HERO issues for aircraft, mobile, and portable emitters that have no fixed location.

In order to determine the calculated safe separation distance for an emitter, specific characteristics, such as average transmitter power, minimum operational frequency, and antenna gain must be identified. Then the MAE or the limits provided in the HERO curves for SUSCEPTIBLE and UNSAFE ORDNANCE (see Figures 1 and 2 in Part 7 of this document) must be identified. By manipulating the Friis free space equation, the above transmitter/antenna characteristics and the desired MAE data can be used to determine the distance at which the EME will affect the susceptibility limit.

10.2 HERO Classifications and HERO Emission Control Procedures. HERO classifications, along with the EME levels in ordnance operating and handling areas serve as the basis for administrative or HERO emission control procedures to preclude HERO. The HERO classifications (that is, HERO SAFE, HERO SUSCEPTIBLE, and HERO UNSAFE) are extremely important in that they immediately identify the level of HERO concern for the warfighter. For example, HERO SAFE ordnance, when encountered, implies that there are no EM concerns so long as the general guidance for ordnance is followed. HERO SUSCEPTIBLE and HERO UNSAFE, when encountered, identify the need to manage HERO, and can be found in the HERO emission control procedures with specific controls to mitigate HERO during specific operations.

HERO controls based on SSDs are more prevalent at installations and activities and are used to manage HERO for aircraft, portable, and mobile emitters that have no fixed location. As described above, the distances can be calculated from transmitter/antenna systems and depends on the frequency, power, antenna gain of the emitter, and the HERO classification of the ordnance.

Installations and activities are typically divided into HERO zones such that when ordnance is present in a particular zone, the necessary emission control procedures can be set. Here, SSDs

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can be used to manage HERO within each of the HERO zones; however, measured data is generally used to help minimize the restrictions placed on transmitting systems since SSDs do not consider the effect the terrain has on the EME. Part 7 provides an example of managing HERO through operational guidance.

11. NOTES

11.1 Intended Use. This handbook may be used by any facility engaged in E3 testing of ordnance for any of the military Services.

11.2 Supersession Information. This handbook supersedes MIL-HDBK-240A, Hazards of Electromagnetic Radiation to Ordnance Test Guide that covered only HERO. This handbook has been broken into Parts to include other facets of E3 testing.

11.3 Subject Term (Keyword) Listing.

E3
Electrically Initiated Device
Electromagnetic
Electromagnetic Compatibility
Electromagnetic Environment
Electromagnetic Environmental Effects
Electromagnetic Radiation Hazards
Electrostatic Discharge
EMC
EME
ESD
Hazards of Electromagnetic Radiation to Ordnance
HERO
Ordnance
Radio Frequency

11.4 Change Notations. Not applicable. This handbook with all of its Parts is a major change and is replacing MIL-HDBK-240A.

11.5 Technical Points of Contact. Requests for additional information or assistance on this standard can be obtained from the following:

Air Force

85th Engineering Installation Squadron (EIS/SCYM)
670 Maltby Hall Drive
Building 7701
Keesler AFB, MS 39534
(228) 377-3920
85eis.scym.1@us.af.mil

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Air Force Safety Center (HQ AFSEC/SEW)
9700 G Avenue, SE
Kirtland AFB, NM 87116
(505) 846-0450
HQAFSCSEW@us.af.mil

Army

US Army Test and Evaluation Command (HQ, ATEC)
6617 Aberdeen Blvd., Building 2202
ATTN: CSTE-OPS
Aberdeen Proving Ground, MD 21005-5001
(443) 861-9664
usarmy.apg.atec.mbx.atec-hq-g3-5@mail.mil

Army Test and Evaluation Command (ATEC)
E3 Test Division (E3 for Missile System Functions)
ATTN: TERT-ECE
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Redstone Arsenal, AL 35898-8052
(256) 313-6970
usarmy.redstone.atec.list.rtc-jocg@mail.mil

Army Test and Evaluation Command (ATEC)
White Sands Test Center (Survivability, Vulnerability and Assessment Directorate)
ATTN: TEWS-SVE
21225 Headquarters Avenue
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(575) 678-6307
usarmy.wsmr.atec.list.datts-sv-e@mail.mil

Marine Corps

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Quantico, VA 22134-6050
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disa.usna.os.list.jsc-j5-e3-engineering@mail.mil

Any information relating to Government contracts should be obtained through contracting officers.

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CONCLUDING MATERIAL

Custodians:
Army – SY
Navy - SH
Air Force – 11
DISA – DC5

Preparing Activity:
DISA (DC5)
(Project EMCS-2022-019)

Review Activities:
Army - AC, AM, AR, AT, AV, CE, CR, GL, MD, MI, PT, TE
Navy – AS, CG, EC, MC, OS, YD
Air Force - 13, 19, 22
DTRA - DS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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